Explore | Expand | Enrich

Agentic AI Workshop

Revision 2

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Agentic AI

Course Overview

This four-day hands-on training program provides foundational knowledge and practical experience in Agentic AI, focusing on building autonomous AI agents that perform tasks independently. Designed for beginners familiar with basic IT concepts and introductory AI terminology, the course progressively guides participants through fundamental AI concepts, agent-based frameworks, prompt engineering, memory-driven reasoning, and agent deployment. Using accessible open-source software and platforms, participants engage in interactive activities, ensuring practical, real-world application of the skills learned.

Course Outcome

- Clearly articulate foundational concepts of AI and differentiate between AI, ML, NLP, and generative models.
- Understand the principles of agentic AI, including autonomy, agent reasoning, and real-world applications.
- Develop basic AI agents using Python, LangChain, and OpenAI's API.
- Craft and optimize effective AI prompts for specific agent behaviors.
- Design advanced agent workflows with memory and decision-making capabilities.
- Deploy AI agents securely & integrate them with external platforms (Slack and APIs).
- Conceptualize, build agent-based solutions, applicable to practical use cases.

Course Prerequisites

- Basic familiarity with Information Technology (IT)
- Understanding of AI fundamentals (basic definitions and concepts)
- No coding expertise required, but comfort with simple scripting is helpful.

Software & Tools

- **Python** (Download)
- **Visual Studio Code** (Download)
- **LangChain** (Python Library, via pip install)

Subscriptions Required

Platform	Subscription Type	URL
OpenAI	Free-tier (API credits provided)	https://platform.openai.com/signup
Google Colab	Free Google Account	https://colab.research.google.com/
Ngrok	Free-tier	https://ngrok.com/signup

High-Level Schedule

Day	Modules (Topics Covered)
Day 1	- Module 1: Introduction to AI & LLMs - Module 2: Prompt Engineering
Day 2	- Module 3: LangChain Fundamentals - Module 4: Retrieval-Augmented Generation (RAG)
Day 3	- Module 5: LangChain Agents & Tools - Module 6: Knowledge Navigation Agent Lab
Day 4	- Module 7: Capstone Project: Education Assistant - Module 8: Deployment with Gradio - Module 9: Presentation & Showcase

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Modules Details

Module 1	Topics Covered
Introduction to AI & LLMs	• What is AI, ML, NLP, GenAI, Agentic AI • Overview of LLMs and GPT • Capabilities and limitations of GenAI • Transition to agentic systems (reasoning, autonomy)
Hands-On Activities: • Setup environment (Google Colab, OpenAI API, LangChain) • Execute first OpenAI API call using a prompt • Group discussion: Examples of reactive vs autonomous AI • Copy-paste prompt templates and run them with OpenAI	
Expected Outcomes: • Participants will understand the distinctions between AI subfields and the significance of LLMs and agentic AI • They will complete their first GPT call and interpret model responses	
Post-Module Capability: • Learner will be able to configure OpenAI and test prompt outputs using API calls in a Colab environment for rapid prototyping.	

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Module 2	Topics Covered
Prompt Engineering	• Prompt structure and syntax • Designing task-specific prompts (instructional, role-based) • Chaining prompts and multi-turn instruction design • Controlling output (temperature, max tokens)
Hands-On Activities: • Modify and run prompt templates • Compare prompt effectiveness (e.g. vague vs structured) • Use role-based prompts (e.g. act as a teacher, summarizer) • Craft a quiz-question generator prompt	
Expected Outcomes: • Participants will gain fluency in prompt design • Learn to control LLM behavior through prompt tweaking • Observe how small changes affect outcomes	
Post-Module Capability: • Design and optimize prompts for diverse task types such as summarization, transformation, and quiz generation	

Module 3	Topics Covered
LangChain Fundamentals	• Overview of LangChain architecture • PromptTemplate and LLMChain • Sequential chains (multi-step processing) • LangChain Memory (ConversationBufferMemory)
Hands-On Activities: • Copy-paste and run LLMChain examples • Build sequential prompt chains (e.g. summarize → rephrase) • Implement a chatbot with conversation memory that remembers user input across turns	
Expected Outcomes: • Participants will be able to build structured workflows using LangChain • Understand how chains manage LLM operations • Understand how memory enables contextual continuity	
Post-Module Capability: • Build modular prompt pipelines using LangChain chains and apply memory to enable context-aware LLM workflows.	

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Module 4	Topics Covered
Retrieval-Augmented Generation (RAG)	• Limitations of LLM internal knowledge • Embedding, vector stores (FAISS concept) • RetrievalQA and document-based answering • Simple custom knowledge integration
Hands-On Activities: • Load sample documents and run queries • Use LangChain's retriever to answer questions from uploaded data • Observe output differences with vs without retrieval	
Expected Outcomes: • Participants will understand how to augment LLMs with external knowledge • Observe improved factuality • Learn to structure queries against a known dataset	
Post-Module Capability: • Integrate external text data sources with LangChain to answer queries using retrieval-augmented generation (RAG).	

Module 5	Topics Covered
LangChain Agents & Tools	• Concept of agents vs chains • Tool design and loading • ReAct paradigm (reasoning + acting) • Agent loop: observe → think → act → reflect
Hands-On Activities: • Build an agent with Wikipedia and Calculator tools • Test complex queries requiring reasoning (e.g. "Square root of year Eiffel Tower was built") • Add a custom tool (e.g. joke generator) to demonstrate extensibility	
Expected Outcomes: • Participants will understand how LLMs can autonomously select and execute tools • Configure agents • Analyze step-by-step reasoning • Learn to add new tools	
Post-Module Capability: • Develop LangChain agents that autonomously select tools and solve multi-step reasoning problems with minimal input.	

Module 6	Topics Covered
Knowledge Navigation Agent Lab	• Task decomposition for multi-step reasoning • Agent-based document querying and synthesis • Summarizing and responding to complex questions • Integrating tools to form logical flows
Hands-On Activities: • Build an agent to fetch Wikipedia content and summarize it • Add logic for generating educational content (e.g. summaries, quiz) • Conduct peer-testing and validation of output quality	
Expected Outcomes: • Participants will develop a working agent that simulates autonomous research behavior • They will combine tools and LLM logic to achieve structured, human-usable output	
Post-Module Capability: • Construct AI agents that conduct basic research by fetching, summarizing, and presenting knowledge using tool integrations.	

Module 7	Topics Covered
Capstone Project: Education Assistant	• Full pipeline agent: retrieval + reasoning + content generation • Structuring prompts for clarity and usefulness • Integrating outputs into an educational flow (summary, quiz, tips) • Deployment preparation
Hands-On Activities: • Plan project tasks (topic selection, tool design, output types) • Code an end-to-end assistant that creates educational content from a topic • Test across topics (e.g. photosynthesis, global warming) • Swap tests with peers for validation	
Expected Outcomes: • Participants will consolidate learning in a visible deliverable – an education-themed AI assistant • They will gain confidence in independently planning and implementing an agentic solution	
Post-Module Capability: • Build and validate a topic-driven education assistant agent that outputs structured learning content from open data.	

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Module 8	Topics Covered
Deployment with n8n	• Basics of n8n.io for quick UIs • Structuring LLM logic as callable functions • Local and Colab-based deployment • Demoing LLM apps securely (without exposing keys)
Hands-On Activities: • Wrap agent in a n8n interface • Run a web app that takes user input and shows AI output • Share the working demo in peer groups • Optional: publish to GitHub	
Expected Outcomes: • Participants will deploy their AI solution into a working interface • They will understand how to package models into accessible web UIs and appreciate the end-user experience	
Post-Module Capability: • Package agent logic into a web-accessible interface using n8n, ready for local demo or GitHub publishing.	

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Module 9	Topics Covered
Presentation & Showcase	• Explain AI solution logic and use-case • Present structured output to non-technical peers • Evaluate agent outputs qualitatively • Discuss future improvements and ethical implications
Hands-On Activities: • Each group presents their project (live demo + explanation) • Conduct real-time topic input tests • Gather peer feedback and ideas • Discuss further exploration areas and share code via GitHub	
Expected Outcomes: • Participants will demonstrate their creations, explain agent design, and gain presentation skills • They will reflect on learnings and be able to explain their AI agent to non-technical stakeholders or future users	
Post-Module Capability: • Present a complete AI solution with reasoning trace and use-case justification to a general or peer audience.	

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Program Format & Logistics

- **Total Duration:** 25 hours (across 4 days)
- **Batch Size:** 30 faculty participants
- **Format:** Instructor-led with extensive hands-on activities
- **Delivery Mode:** Onsite (with pre-configured systems)
- **Prerequisites:** Basic IT background and AI familiarity
- **Tools Used:** OpenAI (free tier), LangChain, Gradio, GitHub, Google Slides/Docs

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